



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(AI&ML)

LLM-Assisted Financial Statement Analysis Tool

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1. Introduction

Financial statements contain important information about a company's financial position and performance.

Manual analysis of statements can require significant time and financial knowledge.

The proposed tool combines data extraction, financial calculations, visualizations and an LLM-based explanation layer.

The goal is to convert raw financial statement data into clear, understandable insights.

2. Problem Statement

Financial information is often presented in lengthy PDF documents or spreadsheets.

Extracting relevant figures manually can be repetitive and error-prone.

Users may find financial ratios and year-to-year changes difficult to interpret.

Traditional tools may show calculations without providing a simple explanation of what the results mean.

Therefore, an integrated tool is needed to automate extraction, analysis and interpretation.

3. Existing System

Manual reading of annual reports and financial statements.

Spreadsheet-based calculations performed separately for different statements.

Users often need to calculate ratios and trends themselves.

Interpretation depends heavily on the user's financial knowledge.

Limited integration between document extraction, numerical analysis and natural-language explanation.

4. Proposed System

Upload financial statements in supported PDF or Excel formats.

Extract relevant financial data into a structured representation.

Calculate selected financial ratios and identify trends.

Use an LLM to explain calculated results in simple language.

Present results through an interactive dashboard and generate a readable report.

5. Objectives

Automate extraction of financial information from uploaded statements.

Reduce repetitive manual work in financial analysis.

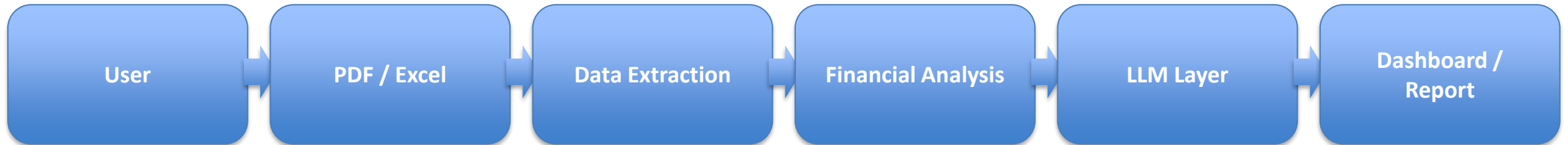
Calculate key financial ratios and compare values across periods.

Generate easy-to-understand AI-assisted explanations of analytical results.

Highlight potential strengths, weaknesses and unusual changes.

Provide a user-friendly dashboard and report output.

6. System Architecture



Input layer: accepts financial statements.

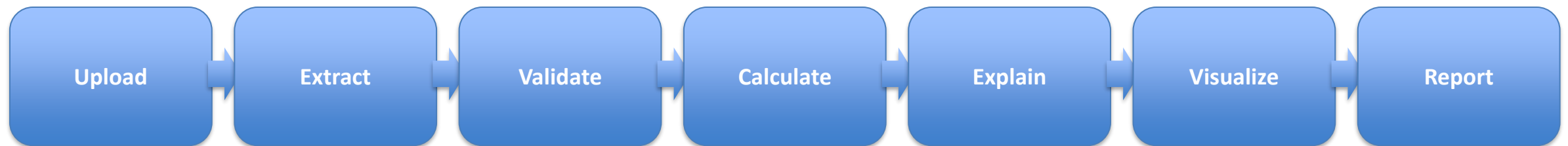
Processing layer: extracts and structures financial data.

Analysis layer: computes ratios and trends.

AI layer: converts analytical outputs into natural-language explanations.

Presentation layer: displays charts, tables, insights and reports.

7. System Workflow



1. User uploads a financial statement.
2. The system extracts required figures and validates the available data.
3. Financial ratios and trends are calculated.
4. The LLM receives structured analytical results and produces explanations.
5. Results are shown on the dashboard and can be included in a report.

8. Technology Stack

Python — core programming and data-processing language.

Flask — lightweight backend web framework.

Pandas — financial data processing and analysis.

OpenPyXL — Excel workbook reading and processing.

PyMuPDF — PDF text extraction.

HTML, CSS and JavaScript — user interface.

Chart.js — dashboard visualizations.

SQLite — lightweight application database.

LLM API — natural-language explanation and insight generation.

9. Financial Analysis Module

Profitability: measures how effectively the business generates profit.

Liquidity: evaluates the ability to meet short-term obligations.

Solvency: examines long-term financial stability and debt position.

Efficiency: evaluates how effectively assets and resources are used.

Trend analysis: compares financial values across reporting periods.

The exact ratio set will be finalized during implementation based on available statement data.

10. LLM-Assisted Insight Generation

The LLM is not intended to replace the numerical calculation engine.

Calculated financial metrics and validated trends are supplied as structured inputs.

The LLM converts these results into concise natural-language explanations.

Possible outputs include explanations of changes, notable trends and areas that may require attention.

Human review remains important because financial decisions should not rely solely on automated interpretation.

11. Dashboard & Report

Summary cards for important financial indicators.

Tables showing extracted values and calculated ratios.

Charts for year-to-year financial trends.

AI-generated explanations beside relevant analytical results.

Downloadable or printable financial analysis report.

Simple interface designed for users with different levels of financial knowledge.

12. Expected Benefits

Faster processing of financial statements.

Reduced repetitive manual calculations.

Centralized view of financial indicators and trends.

Improved accessibility through plain-language explanations.

Better visualization of changes across reporting periods.

Useful foundation for further financial analytics and decision-support features.

13. Future Scope

Support additional financial statement formats and scanned documents.

Improve table extraction and data validation using advanced document AI.

Add benchmarking against industry or peer-company metrics.

Introduce anomaly detection for unusual financial movements.

Add forecasting and scenario analysis.

Support multiple companies and comparative analysis.

Enhance report generation and role-based access.